Return to Play and Decreased Performance After Anterior Cruciate Ligament Reconstruction in National Football League Defensive Players

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Background: Anterior cruciate ligament (ACL) injuries occur commonly in football. Recent work has reported ACL reconstruction (ACLR) as one of several orthopaedic procedures with unfavorable outcomes for professional athletes. The performance impact to defensive players after surgery has not been quantified.

Purpose: To quantify the effect of ACLR on the performance of defensive players by comparing them to a cohort of matched controls as well as to measure the effect of ACLR on athletes' career length in the National Football League (NFL).

Study Design: Case-control and cohort study; Level of evidence, 3.

Methods: Thirty-eight NFL defensive players with a history of ACLR from 2006 to 2012 were identified. For each injured player, a matched control player was identified. Demographic and performance statistics were collected from the online NFL player database. Players who returned after ACLR (n = 23) were compared with players who did not return (n = 15) using *t* tests and chi-squared analyses. Similarly, players who returned after ACLR (n = 23) were compared with their matched controls with *t* tests and chi-squared analyses. Two-way repeated-measures analysis of variance was utilized to test for significant differences between performance before and after the season of the injury for the players in the ACLR group who returned (n = 23) and for their matched controls. Kaplan-Meier analysis was performed to test for differences in the rate of retirement between the groups. For all analyses, *P* values $<.05$ were considered significant.

Results: Approximately 74% (28/38) of athletes who underwent ACLR returned to play at least 1 NFL game, and 61% (23/38) successfully returned to play at least half a season (ie, 8 games). Athletes in the ACLR group who returned retired from the NFL significantly sooner and more often after surgery than their matched controls. In the seasons leading up to their injury, athletes who successfully returned to play started a greater percentage of their games (81%) and made more solo tackles per game (3.44 $\pm$ 1.47) compared with athletes in the ACLR group who did not return to play (54% and 1.82 $\pm$ 1.17, respectively) and compared with healthy control players (52% and 1.77 $\pm$ 1.19, respectively). After the season of surgery, athletes in the ACLR group who returned to play decreased to 57% games started and 2.38 $\pm$ 1.24 solo tackles per game, while their matched controls suffered no significant decreases.

Conclusion: Players who successfully returned were above-average NFL players before their injury but comparatively average after their return.

Keywords: ACL; knee injury; sport injury; NFL

Anterior cruciate ligament (ACL) injuries commonly occur in athletes. These injuries may result from jumping, cutting, pivoting, acceleration/deceleration, collisions, and even celebrations.[4,16] ACL injuries in high-level athletes often result in season-ending tears to the ligament, requiring operative management in which the ACL is reconstructed. Reconstruction must then be followed by rehabilitation.[1,2,5] It is a daunting road to recovery, creating anxiety over the patient's health, athletic performance, and quality of life with no guarantee of return to play.[6,14,17] ACL injuries are prevalent among professional football players, accounting for 16% of reported knee-related injuries.[7] A previous study by Shah et al[17] quantified the rate (63%) and time to return to play (10.8 months) for professional football players after ACL reconstruction (ACLR). Another study reported ACLR as one of several orthopaedic procedures with less than satisfactory outcomes including a lower return-to-play rate and decreased number of games played from baseline upon

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returning.[15] A third quantified the career effect of ACLR on wide receivers and running backs in the National Football League (NFL).[10] However, these studies and others have failed to quantify the unique effect of ACLR on defensive players' performance.[13,15] Defensive players may vary from previously studied running backs, wide receivers, and quarterbacks by body mass index (BMI) and weight, which have been demonstrated as risk factors for an ACL injury.[11] In addition, defensive players execute different strategies and techniques than offensive players. Thus, performance is gauged by metrics that apply specifically to defensive players (eg, tackles/game), and therefore, defensive players should be analyzed separately from offensive players within the context of ACLR outcomes in the NFL. Furthermore, certain positions, namely, defensive backs and linebackers, perform motions such as backpedaling and pivoting that place an additional burden on the reconstructed ACL after surgery that offensive players typically do not perform. The purpose of this study was to quantify the effect of ACLR on the performance of defensive players by comparing them to a cohort of matched controls as well as to measure the effect of ACLR on athletes' career length in the NFL.

METHODS

Study Design

This study was reviewed by the St Vincent's Health System institutional review board and was designated as exempt from requiring review. A list of all NFL injury reports from 2006 to 2012 was generated from publicly available news articles and media reports. The list was filtered for players documented as "out with knee injury." The list was then further filtered by position, limiting the selection to defensive players only. Each injury was then researched further in media and team press releases to assess whether there was an ACL tear and subsequent reconstruction. Players considered for study group enrollment were not excluded based on concomitant injuries (eg, meniscal tears) or previous injuries. Thirty-nine injuries that met the criteria were identified, and these 39 players were included in the study.

Because of the limitations of publicly available sources, concomitant injuries were not recorded or assessed, nor was graft selection. However, based on a survey of orthopaedic surgeons serving NFL organizations, either a hamstring or middle-third patellar tendon autograft for ACLR may be considered the preferred graft choice.[7] Furthermore, additional studies have shown little to no difference in surgical outcomes based on autograft selection or technique.[6] The NFL player database available on the NFL website (http://www.nfl.com/players) was then used to record demographic and anthropometric statistics including age, height, weight, position, years of experience in the NFL before injury, and draft round as well as yearly performance statistics. The position was classified as defensive back, defensive lineman, or linebacker. Yearly performance statistics included games, percentage of games started, solo tackles, assisted tackles, combined tackles (ie, solo tackles plus assisted tackles), quarterback sacks, safeties, number of passes defended from quarterback to receiver, interceptions, total interception return yards, average interception return yards, longest interception return, defensive touchdowns, and forced fumbles. Statistical categories recorded were identical for all defensive players regardless of the defensive position. Performance statistics were collected for 2 seasons before the injury season, the injury season, the return-to-play season, and 2 seasons after the season in which the athlete returned to play. Three seasons of study after ACLR were chosen to ensure an adequate sample of games played after returning to play. Thus, the complete study consisted of data taken over a total of 6 seasons: 3 before surgery and 3 after surgery. After collecting return and performance data on all 39 injured players, it was discovered that 1 player was deceased after a non–football-related accident that occurred during the fall season in which he returned to regular-season NFL play. To avoid categorizing the success of his surgery as a failure based upon his unfortunate but unrelated death or as a success based on an assumption that he would have a successful postinjury career, this player was excluded, leaving 38 total players.

A matched control group of 38 players was created by querying the team roster during the injury season for each player in the ACLR group and then identifying a control player matched on position and total years of prior NFL experience. Controls were selected based on years of NFL experience, as it represents a compilation of similar opportunity between the study athlete and his control. If no control player could be adequately matched (same position, ± 2 years of experience), a control player was selected from another team from the same season. This selection method also minimized variations in the opposing team's offensive performance by comparing players in the same season and against the same opponents. The exclusion criteria for selecting a control player were reported history of ACL injuries at any level of play and any injury that resulted in greater than 3 weeks on the injured reserve list during the "match season" and 2 preceding seasons. A match season was defined for each matched pair of players as the season in which the player who underwent ACLR sustained his injury. Considerable research was performed using media outlets, the NFL website, and team websites, among others, to ensure that no matched control athlete had previously undergone ACLR during his professional or collegiate career. Demographic and performance statistics of control players were gathered in an identical manner using the protocol described above. This method of matching control players was tested by comparing demographic, anthropometric, and performance statistics between the 2 groups. There were no significant differences found, and it was concluded that the 38 controls had similar anthropometric, playing history, and performance statistics as the players who underwent ACLR.

Statistical Analysis

Statistical analysis was performed using SAS 9.4 (SAS Institute). Demographic and performance statistics before surgery were compared between 2 subgroups of the

TABLE 1
Demographics and Performance of ACLR Group Before Surgery[a]

	Returned to Play (n = 23)	Did Not Return to Play (n = 15)	*P* Value	All Players in ACLR Group (n = 38)
Demographics				
Age, y	27.3 ± 3.1	28.0 ± 3.2	.54	27.6 ± 3.1
Height, m	1.86 ± 0.05	1.86 ± 0.04	.92	1.86 ± 0.04
Weight, kg	100.8 ± 12.0	117.0 ± 26.5	**.039**	107.2 ± 20.4
Body mass index, kg/m^2	28.9 ± 2.5	33.5 ± 6.9	**.024**	30.7 ± 5.2
Surgery season, n (%)			.75	
Spring	0 (0)	0 (0)		0 (0)
Summer	3 (13)	3 (20)		6 (16)
Fall	15 (65)	8 (53)		23 (61)
Winter	5 (22)	4 (27)		9 (24)
Draft round, n (%)			.30	
During first 4 rounds	17 (74)	8 (53)		25 (66)
After first 4 rounds	6 (26)	7 (47)		13 (34)
Position, n (%)			.16	
Defensive lineman	3 (13)	6 (40)		9 (24)
Linebacker	8 (35)	4 (27)		12 (31)
Defensive back	12 (52)	5 (33)		17 (45)
NFL experience at injury, y	4.6 ± 2.8	4.8 ± 3.0	.81	4.7 ± 2.9
Performance				
Games started, %	80.9 ± 28.5	52.3 ± 34.9	**.014**	69.6 ± 33.9
Combined tackles/game	4.28 ± 1.99	2.42 ± 1.35	**.002**	3.55 ± 1.97
Solo tackles/game	3.44 ± 1.47	1.77 ± 1.19	**<.001**	2.78 ± 1.58
Assisted tackles/game	0.85 ± 0.61	0.61 ± 0.34	.13	0.75 ± 0.53
Sacks/game	0.13 ± 0.19	0.05 ± 0.07	.12	0.10 ± 0.16
Forced fumbles/game	0.065 ± 0.057	0.034 ± 0.032	**.036**	0.053 ± 0.051
Passes defended/game	0.41 ± 0.31	0.12 ± 0.23	**.002**	0.30 ± 0.31
Interceptions/game	0.095 ± 0.086	0.019 ± 0.044	**.001**	0.065 ± 0.081
Safeties/game	0.001 ± 0.007	0	.33	<0.001 ± 0.005
Defensive touchdowns/game	0.007 ± 0.015	0	**.046**	0.004 ± 0.012

[a]Data are provided as mean ± SD unless otherwise indicated. A player was considered to have successfully returned if he played in at least 8 National Football League (NFL) games after anterior cruciate ligament reconstruction (ACLR). Performance before surgery includes data for the season of surgery as well as the 2 preceding seasons. Significant *P* values (<.05) in bold.

ACLR group: those who successfully returned to play and those who did not. Successful return to play was defined as returning to and participating in at least 8 regular-season NFL games (ie, half a season) after ACLR. The percentage of players who returned to play for at least 1 game and the date of the last game played were recorded through the conclusion of the 2015 season. Demographic and performance statistics before surgery were also compared among the athletes in the ACLR group who did return to play and their matched controls. These comparisons were made using independent Student *t* tests and chi-squared tests as appropriate.

Data from before and after the injury were compared between the athletes in the ACLR group who returned and their matched controls. Data were analyzed using 2-way (before vs after surgery, ACLR vs control) repeated-measures (participant) analysis of variance (ANOVA). Two-way repeated-measures ANOVA was also performed on the 3 subgroups of positions: defensive linemen, linebackers, and defensive backs. For each significant ANOVA, differences over time (before vs after surgery) were then analyzed for each group (ACLR or control) using a post hoc paired *t* test. NFL survival rates were compared among the players in the ACLR group who returned to sport and their matched controls, with retirement as an endpoint using Kaplan-Meier survival analysis and a Wilcoxon test to assess for a difference between the rates. A difference was considered statistically significant if the *P* value was <.05.

RESULTS

Twenty-eight (74%) of the 38 injured defensive players returned to play at least 1 NFL game, but only 61% (23/38) successfully returned to play at least half a season (8 regular-season games). A comparison of those who did and did not successfully return after ACLR is shown in Table 1. Players who returned did so a mean of 47.0 ± 16.2 weeks (range, 29-101 weeks) after surgery. While there were no significant differences in age, height, or position played at the time of surgery, the athletes in the ACLR group who successfully returned had less weight and a lower BMI. Those who returned to play had also started in a greater percentage of their games before surgery than those who did not successfully return. In addition, players

TABLE 2
Demographics of ACLR Group With Successful Return and Control Group[a]

	ACLR Group With Successful Return (n = 23)	Control Group (n = 23)	*P* Value
Age, y	27.3 ± 3.1	27.2 ± 3.2	.90
Height, m	1.86 ± 0.05	1.86 ± 0.06	.77
Weight, kg	100.8 ± 12.0	104.4 ± 17.6	.41
Body mass index, kg/m^2	28.9 ± 2.5	30.1 ± 3.7	.23
Draft round, n (%)			.53
During first 4 rounds	17 (74)	14 (61)	
After first 4 rounds	6 (26)	9 (39)	
Position, n (%)			—
Defensive lineman	3 (13)	3 (13)	
Linebacker	8 (35)	8 (35)	
Defensive back	12 (52)	12 (52)	
NFL experience at injury, y	4.6 ± 2.8	4.6 ± 3.2	>.99

[a]Data are provided as mean ± SD unless otherwise indicated. A player was considered to have successfully returned if he played in at least 8 National Football League (NFL) games after anterior cruciate ligament reconstruction (ACLR).

TABLE 3
Performance Before Injury and After Surgery[a]

	ACLR Group With Successful Return (n = 23)		Control Group (n = 23)		
	Before	After	Before	After	*P* Value
Games started, %	80.9 ± 28.5[b,c]	57.0 ± 33.0[b]	60.5 ± 34.1[c]	61.9 ± 37.2	**.004**
Combined tackles/game	4.28 ± 1.99[b]	3.17 ± 1.74[b]	3.27 ± 1.56	3.07 ± 1.66	**.013**
Solo tackles/game	3.44 ± 1.47[b,c]	2.38 ± 1.24[b]	2.50 ± 1.16[c]	2.28 ± 1.20	**.005**
Assisted tackles/game	0.85 ± 0.61	0.80 ± 0.65	0.73 ± 0.46	0.79 ± 0.56	.30
Sacks/game	0.13 ± 0.19	0.09 ± 0.12	0.10 ± 0.15	0.08 ± 0.15	.48
Forced fumbles/game	0.065 ± 0.057	0.056 ± 0.054	0.055 ± 0.082	0.048 ± 0.055	.91
Passes defended/game	0.41 ± 0.31[b]	0.31 ± 0.27[b]	0.33 ± 0.31	0.28 ± 0.26	.37
Interceptions/game	0.095 ± 0.086[b]	0.047 ± 0.045[b]	0.057 ± 0.069	0.073 ± 0.091	**.005**
Safeties/game	0.001 ± 0.007	0.000 ± 0.000	0.008 ± 0.032	0.004 ± 0.011	.76
Defensive touchdowns/game	0.007 ± 0.015	0.006 ± 0.015	0.007 ± 0.017	0.005 ± 0.010	.64

[a]Data are provided as least squares mean ± SD. For each matched pair of players, "before" data include the season of the injury and 2 preceding seasons, and "after" data include the 3 seasons after the injury. A player was considered to have successfully returned if he played in at least 8 National Football League games after anterior cruciate ligament reconstruction (ACLR). Significant *P* values (<.05) in bold.
[b]Significant within-participant difference (before vs after) in group.
[c]Significant preinjury difference (ACLR vs control) between groups.

who successfully returned generally performed better leading up to surgery than those who did not return, as seen by significantly more combined tackles, solo tackles, forced fumbles, passes defended, interceptions, and defensive touchdowns per game.

While the demographics and playing history of the players in the ACLR group who returned were not different from their controls (Table 2), the preinjury performance of those who successfully returned to the NFL was compared with that of their matched controls and was found to be significantly better (Table 3). Specifically, the ACLR group had a higher percentage of games started and had more solo tackles per game than the control group.

Of particular interest to the purpose of this study was the comparison of performance before the injury and after surgery between the ACLR group and control group (Table 3). Compared with their performance before the match date, the controls had no significant changes in performance after the match season, whereas the athletes in the ACLR group who successfully returned to play experienced a number of reductions in performance measurements. The percentage of games started, combined tackles per game, solo tackles per game, and interceptions per game all decreased significantly for the ACLR group after their return. Control players experienced no significant declines in on-field performance during the 3 years after their matched athlete underwent ACLR. In addition to decreased in-game performance, players who successfully returned after ACLR appeared to retire more quickly and more often after surgery than the control players, although this trend did not reach statistical significance ($P = .064$); full survival data can be seen in Figure 1.

Position-specific results were analyzed for percentage of games started, solo tackles per game, and interceptions per game (Table 4), as these were the variables that showed

TABLE 4
Significant Changes for Specific Positions[a]

	ACLR Group With Successful Return (n = 23)		Control Group (n = 23)		
	Before	After	Before	After	*P* Value
Games started, %					
Defensive linemen	88.2 ± 20.2	57.3 ± 42.7	76.9 ± 33.7	90.8 ± 10.1	.17
Linebackers	84.6 ± 23.1	69.0 ± 30.9	62.89 ± 32.1	53.6 ± 33.9	.52
Defensive backs	76.6 ± 34.3[b]	49.3 ± 33.3[b]	54.8 ± 36.9	60.2 ± 41.7	**.022**
Solo tackles/game					
Defensive linemen	2.21 ± 0.41	1.07 ± 0.54	1.88 ± 0.55	1.43 ± 0.45	.12
Linebackers	4.07 ± 1.60[b]	2.96 ± 1.25[b]	2.70 ± 1.47	2.39 ± 0.96	.11
Defensive backs	3.32 ± 1.36[b]	2.32 ± 1.13[b]	2.53 ± 1.06	2.41 ± 1.42	.060
Interceptions/game					
Defensive linemen	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.060	0.009 ± 0.015	.14
Linebackers	0.070 ± 0.056	0.054 ± 0.047	0.005 ± 0.010	0.023 ± 0.028	**.039**
Defensive backs	0.130 ± 0.091[b]	0.055 ± 0.045[b]	0.100 ± 0.071	0.120 ± 0.110	**.013**

[a]Data are provided as least squares mean ± SD. For each matched pair of players, "before" data include the season of the injury and 2 preceding seasons, and "after" data include the 3 seasons after the injury. A player was considered to have successfully returned if he played in at least 8 National Football League games after anterior cruciate ligament reconstruction (ACLR). Significant *P* values (<.05) in bold.
[b]Significant within-participant difference (before vs after) in group.

significant differences for the overall cohort. Combined tackles per game also showed changes for the cohort of players who underwent ACLR, but this variable was omitted from the position-specific analysis, as it was assumed to be an effect of the change in solo tackles per game. Defensive backs within the ACLR group suffered significant decreases in the percentage of games started and interceptions per game. Linebackers showed significant decreases in interceptions per game. There were only 3 players in the group of defensive linemen who returned to play after ACLR, so the apparent decreases in the percentage of games started and solo tackles per game were not statistically significant.

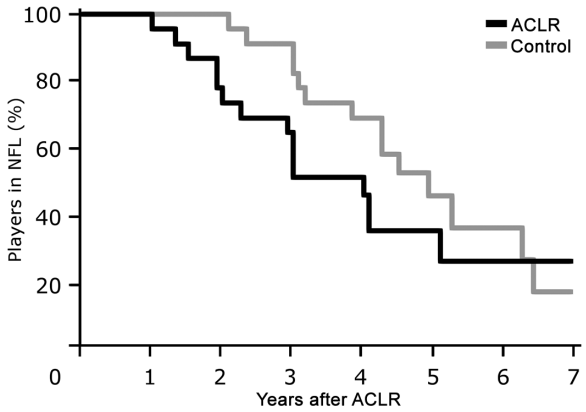

Figure 1. Survival in the National Football League in the anterior cruciate ligament reconstruction (ACLR) group versus control group. Wilcoxon group difference: *P* = .06.

DISCUSSION

All players who reach the NFL have exceptional talent and physical attributes. That being said, some NFL players are better than others. The athletes in the ACLR group who returned to play appeared to be the more elite NFL players before their knee injury, as they had been named the starter at their position more often and had better performance before their ACL injury compared with a matched control group of healthy, typical NFL players. They had also started more often in games and had better performance than the athletes in the ACLR group unable to return to play. However, the players who returned after ACLR did so with significantly diminished performance compared with their own preinjury levels. Therefore, it appears that those in the ACLR group who successfully returned were elite NFL players before their ACL injury but average NFL players after ACLR. This can be seen in Figures 2 and 3, which show the ACLR group that successfully returned as significantly better than the control group before the injury but similar to the control group after surgery. It is possible that the decrease in tackles and interceptions per game may be a function of decreased overall playing time (as players in the ACLR group started in a lower percentage of their games played compared with their preinjury levels); nevertheless, the fact remains that elite athletes are able to return after ACLR but do so at a level similar to average NFL players.

Previous work has linked an athlete's likelihood of returning after ACLR to his draft round, with earlier draft-round picks being more likely to return.[17] The proposed explanation for this relationship was that higher-round picks represented a greater financial investment to the team, and therefore, management, coaching staff, and trainers may have been more incentivized to return these injured athletes to the playing field than their counterparts who were selected later in the draft. This phenomenon was also recognized among athletes returning after lateral

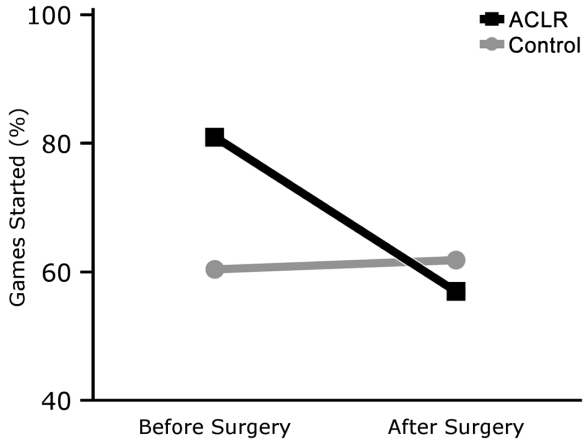

Figure 2. Percentage of games started among athletes who returned to play in the anterior cruciate ligament reconstruction (ACLR) group and their matched controls. Two-way analysis of variance: $P = .004$. Significant preinjury difference between groups ($P = .004$). Significant decline within ACLR group after return to play ($P < .001$).

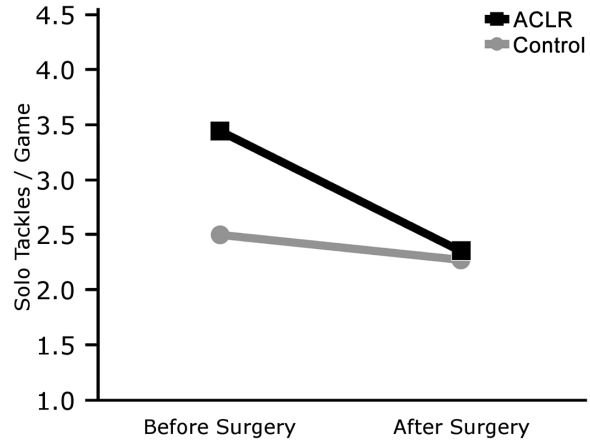

Figure 3. Solo tackles per game among athletes who returned to play in the anterior cruciate ligament reconstruction (ACLR) group and their matched controls. Two-way analysis of variance: $P = .005$. Significant preinjury difference between groups ($P = .022$). Significant decline within ACLR group after return to play ($P < .001$).

meniscectomy, and a similar explanation was offered.[3] However, on-field performance was not included in either of these studies, as it was in the current study, and should be considered as a significant contributor. In the present study, players in the ACLR group who did not return were, indeed, average NFL athletes before their injuries; perhaps their talent after surgery deteriorated to below that which would be required to continue to play at the NFL level, explaining why they were unable to return.

While the players in the ACLR group who did and did not return had similar ages and heights, those who returned had approximately 16 kg less body weight. We believe that this is because of the ACLR group that successfully returned having a higher percentage of defensive backs and a lower percentage of defensive linemen, although these different position distributions did not reach statistical significance. Alternatively, increased weight and BMI are proven risk factors for ACL injuries, and therefore, a lower BMI and weight may prove to be positive prognostic factors for recovery and postoperative performance with further investigation.[11] Although the current study did not have large enough sample sizes in the subgroups to show statistical significance, there seemed to be a better return at defensive positions with smaller, faster players. Based upon the data in Table 1, 71% of the defensive backs, 67% of the linebackers, and 33% of the linemen returned to play. Assuming these differences are true, the disparities in return rates may be caused by a combination of variations in body weight, physiology, athleticism, and biomechanical demands of various defensive positions.

Previous reports also investigated performance changes after ACLR. Carey et al[10] used a measure called a power rating (total offensive yards/(10 + 6 × total touchdowns)) to compare the performance of running backs and wide receivers in the NFL before and after ACLR and noted a significant drop in performance among the 79% of players in the ACLR group who returned after surgery compared with healthy controls. In a separate study analyzing all players (offensive and defensive) who underwent ACLR, Mai et al[15] used a scoring system incorporating offensive and defensive metrics for each respective player to arrive at a total performance score for each player before and after ACLR. The authors described a significant decrease in performance in the 2 years after the injury compared with the preinjury performance of the 82% of players who returned after ACLR. However, this study lacked a control group to account for any declines in performance that may occur naturally as an athlete continues to play such a physical sport. Despite differences in methodology, study populations, and measured outcomes, the findings of the current study are in agreement with these 2 previous studies; many athletes return to sport after ACLR but do so with significant declines in on-field performance when compared with their preinjury levels.

The issue of return to play and competitive performance after ACLR often arises in preoperative counseling. This report, along with previous literature, may be used to provide athletes with expectations regarding rehabilitation duration, approximate date of return to play, chances of returning to play, and realistic expectations of performance and level of competition after returning to play.

Limitations and Strengths

This study is not without limitations. It was a retrospective study whose participant population was sourced from a list of injuries generated from the NFL website, lay press, and media reports, not official communications from medical spokespersons for teams or athletes, and therefore may not be entirely inclusive of all ACL injuries sustained by NFL defensive players during this time period. However, we have no reason to believe that any incidental omission

of an injured player from our list of ACL injuries occurred in a way that would have introduced any systematic bias into the study. Another weakness is the lack of access to medical data concerning the athletes. Previous work has shown that concomitant meniscal injuries and treatment have a significant effect on an athlete's career length compared with isolated ACLR[8]; however, information contained in operative notes including reconstruction techniques and concomitant injuries was not available for review, and rehabilitation protocols were likely not standardized. Furthermore, while efforts were made to ensure that the control group had never undergone previous ACLR, it is possible that, without access to these records, an athlete may have undergone surgery at a time when it was unlikely to be reported in the media, such as in high school or at a smaller university. Although the authors believe that 38 NFL defensive players comprise an adequate sample size to draw general conclusions concerning defensive players, position-specific performance assessments contained small sample sizes. However, this study has comparable statistical power to previous studies that also investigated outcomes after ACLR in limited populations.[9,12] The case-control design is a strength of this study. The matched control group allowed for an accurate assessment of whether a change in performance for the ACLR group was unusual for NFL defensive players. Finally, the generalizability of this study is limited to professional American football players. It is unclear whether the same declines in postsurgical performance would be noted among football players at other levels of competition or among athletes in other sports.

CONCLUSION

Only about 61% of NFL defensive players returned to a competitive level of play after ACLR, with only 26% remaining active in the NFL 3 seasons after surgery. Those who returned were above-average NFL players before their injury but relatively average after their return.

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